

Analyzing factors influencing defect-based conditions for sewer pipes using Bayesian networks

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ABSTRACT

The failure and condition of sewer pipes are usually influenced by various uncertainties and factors. Therefore, it is vital to evaluate the defect-based condition of sewer pipes for maintenance and failure prevention. In contrast to relying on the existing domain knowledge, this paper proposes a data-driven Bayesian network (BN) model to analyze the impacts of different influence factors (IFs) on the sewer pipes' defect-based condition based on the developed database about the Hong Kong sewer network. Specifically, fourteen IFs are collected to develop a multi-source integrated database that incorporates pipe physical, environment, and climate-related factors. Then, the structure of the BN model is learned by Bayesian searching algorithm, and the reliability of the proposed model is evaluated using sensitivity analysis methods. Moreover, special scenarios are assumed to explore possible configurations of IFs. The results reveal that age, diameter, population and soil type are the top four IFs affecting the condition of sewer pipes, among which pipes with some characteristics need to be closely monitored, such as pipe age greater than 50 years, diameter less than 200 mm, location with a population density less than 15,000, and location in fill or granitic rocks. This paper improves sewer pipe management by integrating a comprehensive database and enabling reliable pipe condition inferences. The insights provide practical suggestions for the sewer pipe layout, risk analysis, and maintenance strategy formulation. It is a precious tool for authorities to enhance the safety and efficiency of sewer system management.

1. Introduction

The sewer system is a vital component of urban infrastructure assets, playing a significant role in preventing waterborne diseases and flooding by safely and effectively conveying sewerage and surface runoff to treatment facilities [1,2]. Hong Kong sewer network (HKSN) spans 4700 km, with an average age of exceeding 30 years; notably, 15 % of these pipes are over 50 years old [3,4]. A batch of sewer pipes approaching their end of service life has experienced an expected increase in structural and operational failures, where structural failure refers to the physical damage to sewer and operational failure denotes a reduction in hydraulic capacity of sewer pipes [5,6]. Sewer pipes with different characteristics would experience significant deterioration of condition due to physical, environmental and operational factors [7,8]. In such cases, not only will the reliability of the sewer system be reduced, but the service capacity of this infrastructure system will also be weakened [9,10]. Therefore, in the absence of an in-depth and comprehensive investigation into the characteristics and interactions among

defect-related factors, the likelihood of sewer failures is expected to rise. Such failures would have detrimental socio-economic-environmental consequences, such as overflow and flooding, water quality degradation, hazardous pollution to the surrounding environment, damage to road, threats to public health and quality of life for nearby communities [11–14].

It is vital to study the causes that affect the condition of sewer pipes, and numerous studies have clustered the influence factors (IFs) as physical, operational and environmental [5,7,15,16]. Some have identified that pipe physical characteristics play critical roles in influencing pipe condition deterioration, such as pipe age, material, and slope [1,17]. Moreover, other factors related to soil properties and climate parameters have received sizeable attention, like soil corrosivity, rainfall and humidity [17–20]. However, existing literature has disagreements regarding the impacts of such factors on sewer pipe condition. As a commonly used technology for detecting and monitoring sewer defects, closed-circuit television (CCTV) has developed a standard interpretation protocol for sewer inspection [21]. Considering that, it is a better

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strategy to build an aggregated evaluation of the pipe's condition by accounting for the frequency and types of defects as well as some grading indicators instead of separate models.

In recent decades, numerous researchers have concentrated on condition evaluation and risk assessment of sewer pipes. Physics-based models are adept at leveraging the physical characteristics and external mechanical properties to determine the pipe conditions, and examples include power function models, linear function models, etc., [7,22]. Although physics-based model is more robust from a scientific perspective, the complexity of deterioration processes in sewer systems still makes it challenging to simulate similar pipe conditions uniformly. In addition, due to the scarcity and difficulty of data organization, the physics-based model is considered too simple to truly reflect the condition deterioration process of sewage pipes [7]. Regarding this, some synthetic methods, such as analytic hierarchy process (AHP), fuzzy logic, and artificial neural network (ANN), have been gradually introduced for the condition assessment of sewer pipes [23–25]. However, apart from the subjectivity that expert opinions may introduce, these methods are still limited when it comes to the dynamic evaluation of complex systems because changes in the involved variables may lead to recalculation of all related parameters, which will greatly reduce the performance of the model. Under this circumstance, the Bayesian network (BN) demonstrates a significant potential due to its flexibility to extrapolate interconnections and uncertainties among variables within a model efficiently [26–29]. As a theory combining graph and probability, the BN model can depict the true causal relationships between the involved variables in visual and graphical ways. Meanwhile, the BN is capable of both forward and backward reasoning, during which it can absorb new information to update the inference results and process the missing data [30,31]. Thus, the BN can be utilized for prediction and fault diagnosis, and it has demonstrated effective in various research fields including pipe systems. For example, Wu et al. [32] utilized the BN model to investigate the accident probability of natural gas pipe networks incorporating expert knowledge. Tang et al. [33] compared the construction of BN structure using automatic learning versus manual methods based on a literature review and exploratory data analysis. Zhang and Weng [34] developed a knowledge-based BN model to analyze failure frequency and leakage size of urban buried gas pipes based on pipe characteristics and historical data. Zhou et al. [35] proposed a risk assessment method based on a BN and Dempster-Shafer evidence theory to evaluate complicated sewer pipe accidents in a utility tunnel. Hassan et al. [36] explored the failure probability of cross-country oil pipe systems by developing the BN model, where they collected the expert opinion, historical cases and considered the interaction between several parameters, like human factors, mechanical factors and natural hazards. Obviously, BN has gained increasing prominence in failure analysis due to its advantages of non-parametric nature and ability to implement probabilistic inference easily and efficiently for each failure outcome by considering all input variables [33, 37].

This study aims to develop data-driven BN models to evaluate the impacts of various IFs on the defect-based conditions for sewer pipes. Firstly, a thorough literature review is conducted to identify key IFs associated with the worsening of sewer pipe condition. Secondly, taking HKS as a pilot study, we develop a multi-source integrated database by collecting data on physical, environmental and climate-related factors as well as defect records from CCTV reports. Next, a data-driven BN model is established through the score-based Bayesian searching algorithm to learn the interactions between IFs and sewer pipe conditions. Finally, sensitivity analysis and model validation are applied to evaluate the feasibility and reliability. Our study will contribute to the sewer management system: (1) it proposes an aggregated index of pipe's condition which combines CCTV-reported defect type and frequency with different grading indicators; (2) it identifies a comprehensive list of IFs leading to change in sewer pipe's defect-based condition; and (3) it develops a data-driven BN model which could provide a superior way to reveal the

impacts of different IFs on sewer pipe's condition, as well as their unexplored interrelationships. This study could provide crucial information for utilities to identify the main causes of sewer failure, prioritize maintenance actions and allocate resources efficiently. This research could optimize sewer pipe failure analysis, improve wastewater infrastructure protection and promote environmental preservation.

The remainder of this paper is structured as follows: Section 2 describes the research methodology of data preparation and the data-driven BN model development that includes BN structure learning, sensitivity analysis and model validation. Section 3 presents a case study showing the content of IFs identification, integrated database and the result of BN model development. Section 4 expounds on the impacts of different IFs on the defect-based conditions of sewer pipes through the sensitivity analysis of the proposed BN model and performs scenario analysis to simulate potential situations. Finally, Section 5 illustrates the concluding remarks of this study.

2. Methodology

To develop the data-driven Bayesian network (BN) model for analyzing the defect-based condition of sewer pipes, a conceptual methodology is proposed, consisting of data preparation, model development, sensitivity analysis and model validation, as shown in Fig. 1. After the data preparation, the data-driven BN model can be constructed based on the integrated database to investigate the impacts of different IFs on the defect-based condition of sewer pipes, during which the structural learning of the BN model is performed through the score-based Bayesian searching algorithms. Finally, sensitivity analysis and model validation are conducted to evaluate the developed BN model.

2.1. Data preparation

2.1.1. IF identification from literature

IF identification, as the first step in data preparation, can avoid a large amount of irrelevant and useless information in the database for defect-based condition analysis of sewer pipes. Relevant literature is the most representative reference for determining IFs, thus, the whole process of IFs identification involves three steps: (1) online database retrieval; (2) retrieval result screening and refining; and (3) IF determination and analysis. IFs which are identified through scientific steps can provide a theoretical support for subsequent data collection and processing.

2.1.2. Data collection

Data collection is a crucial step before developing data-driven models. Although most data from public agencies is gradually becoming accessible, there are still challenges in terms of data integration due to the differences in storage formats and agents [38]. This study endeavours to establish a multi-source integrated database for assessing sewer pipe's defect-based conditions, where the historical IFs data, pipe inspection records, CCTV reports from different departmental organizations will be collected. Departmental organizations in Hong Kong that can provide such data include Drainage Services Department (DSD), Hong Kong Observatory (HKO), etc.

2.1.3. Data processing

The collected data shall be further processed to improve its quality and usability, which consists of data integration, data cleansing, and determination of IFs and the target variable. Data integration requires combining and matching the data in the database according to the same benchmark. In this study, data integration can be carried out based on the unique pipe ID. To ensure the accuracy and consistency, data cleansing can be applied such as deleting duplicate records, supplementing missing data, and correcting inconsistent data in format. After that, the developed database in this study also needs to be organized into a form that BN can use through data discretization and state

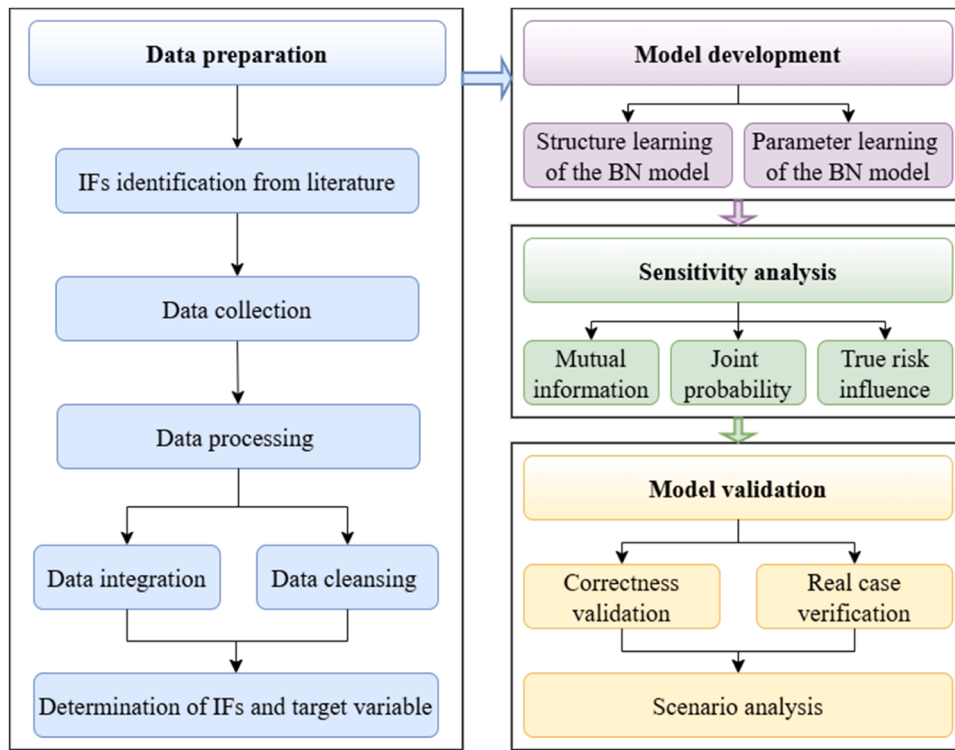


Fig. 1. The flowchart of the adopted methodology in this study.

simplification, where the most important issue is the determination of the target variable "pipe condition".

To explore the impacts of IFs on the condition of sewer pipes, it is important to adopt a normative rating system to represent the level of defect-based condition for each sewer pipe. Commonly used grading indicators include internal condition grade (ICG), service / operational condition grade (SCG) and structural performance grade (SPG); the values of ICG and SCG are stipulated by structural and operational defects, respectively, while SPG is another grading indicator based on ICG and adjusted with supplementary information. The values of these indicators cannot be directly obtained from raw survey but need to be manually extrapolated based on the observed defect from CCTV reports. The standards of defect classification and grading are slightly different among countries and regions [39]. This study adopts the Hong Kong Conduit Condition Evaluation Codes [40] as the protocol for the entire analysis. Table 1 presents the ICG and SCG for some typical defects of sewer pipe. It can be observed that the values of ICG and SCG range from 1 to 5, where the larger the value, the more severe the defect. As ICG can only rate the structural condition based on internal conditions without considering the surrounding environment, the assessment may be inaccurate in some marginal cases, where SPG can provide more comprehensiveness. There are hills and slopes in Hong Kong, meanwhile, most sewer pipes are installed in the early years. Therefore, we mainly consider whether the sewer is constructed on slope and year of installation in pipe condition determination [40]. The calculation equation of SPG is as follows.

$$SPG = ICG + V \quad (1)$$

where V is the variable adjusting ICG by considering additional information, and the value of V is determined through Table 2. Eventually, the target variable (i.e., pipe condition) can be inferred by taking the maximum value among ICG, SCG and SPG, and divided into six levels ranging from 0 to 5, namely perfect, excellent, good, fair, poor and critical. The whole process on how to determine pipe's defect-based condition is shown in Fig. 2.

Table 1

ICG and SCG for different types of defects, adopted from Hong Kong conduit condition evaluation codes.

Defect category	Defect type	Description	Grade
Structural	Crack	Crack longitudinal	1
		Crack circumferential	1
		Crack multiple	2
	Fracture	Fracture longitudinal	3
		Fracture circumferential	3
		Fracture multiple	4
	Broken	Broken pipework	4
	Hole	Hole above/below	4
	Collapse	Collapse pipe/brickwork/masonry	5
	Deformation	Deformation ($\leq 5\%$)	3
		Deformation (between 5 % and 25 %)	4
	Joint	Joint displaced	2-3
		Open joint	1-2
	Surface damage	Surface spalling	1-3
		Surface wear	1-3
	Junction & connection	Connection defective	4
	Repair defective	Lining defect	4
Service (Operational)	Deposits	Deposits attached/settled	1-5
	Obstacles	Obstacles	5
	Encrustation	Encrustation light	2
		Encrustation medium	4
		Encrustation heavy	5
	Root	Root fine	2
		Root tap	3
		Root mass	2-5
	Infiltration	Infiltration seeping	2
		Infiltration dripping	2
		Infiltration running	4
		Infiltration gushing	5

Table 2

Supplementary information of SPG based on ICG, adopted from Hong Kong conduit condition evaluation codes.

Location of pipe	Years of installation		
	<30 years	Between 30 and 50 years	>50 years
On slope	+1	+2	+2
Not on slope	0	+1	+2

2.2. Model development

BN is a graphical reasoning method, which can be generated by subjective or objective methods. It can be described as a directed acyclic graph (DAG) that includes parent nodes and child nodes, meanwhile, dependencies and causal relationships between nodes are displayed by graphic links and conditional probability tables (CPTs) [41]. BN has been proven to be an effective tool in conducting risk assessment and decision-making, and widely applied in accident prediction, risk analysis, failure diagnosis and other fields [42]. Bayesian theory is the basis for BNs, in which the condition probabilities can be calculated as follows.

$$P(X|Y) = \frac{P(Y|X) * P(X)}{P(Y)} \quad (2)$$

where $P(X)$ and $P(Y)$ represent the prior probabilities of the event X and Y , respectively; $P(Y|X)$ presents the probability of event Y when event X happens; and $P(X|Y)$ refers to the probability of event X when event Y occurs. BN achieves outstanding performances in data update capabilities, which provides a convenient method for systematic modeling. The posterior probability of an event can be obtained by updating the prior probability of new evidence.

$$P(X|e) = \frac{P(e|X) * P(X)}{P(e)} = \frac{P(e|X) * P(X)}{\sum_{i=1}^n P(e|Xi)P(Xi)} \quad (3)$$

where $P(X|e)$ presents the posterior probability of event X based on giving evidence e ; $P(e|X)$ is the probability of evidence e when event X happens; and $\sum_{i=1}^n P(e|Xi)P(Xi)$ represents the joint probability distribution of evidence e . BN demonstrates great flexibility in calculating joint probability distribution, and its ability to evidence update constitutes a powerful combination of prediction, diagnosis, and explanation.

When performing BN analysis, two consecutive steps are crucial: structural learning and parameter learning, in which the structure learning process can be conducted in two ways containing subjective expert knowledge and objective data-driven algorithms [43]. Subjective information, such as expert experience, judgement, and/or common sense, is relatively easy to be collected. However, this process is time-consuming and unreliable since it may involve personal bias. In comparison, data-driven approaches using historical data can greatly increase the reliability and robustness [28,44]. Therefore, a score-based Bayesian searching algorithms is employed to develop BN models, among which scoring functions (SFs) are used to determine the best DAG after searching all possible options. Typically, commonly-used algorithms include K2, Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), in which BIC outperforms the other algorithms while simulating large samples, hence BIC is applied to the process of data simulation. Given a dataset D containing N samples, each sample contains n variables, the BIC of a BN structure can be expressed as follows [45]:

$$BIC(D) = \hat{L}(D) - \frac{1}{2} \log(N) |\theta| \quad (4)$$

where $|\theta|$ presents the number of parameters in the model; and $\hat{L}$ represents the maximum value of the log-likelihood function of the model that can be defined as follows:

$$LL(D) = \sum_{i=1}^n \sum_{k=1}^{r_i} \sum_{j=1}^{q_i} N_{ijk} \ln \frac{N_{ijk}}{N_{ij}} \quad (5)$$

where N_{ij} represents the number of samples that the variable i with its parents j ; N_{ijk} is the number of instances when the variable i is with its parents j , in which the variable i is in the state k ; and the number of different states of the variable i and the number of its possible parent variable are represented by r_i and q_i , respectively. After the BN structure is determined, the parameter learning can be performed based on the

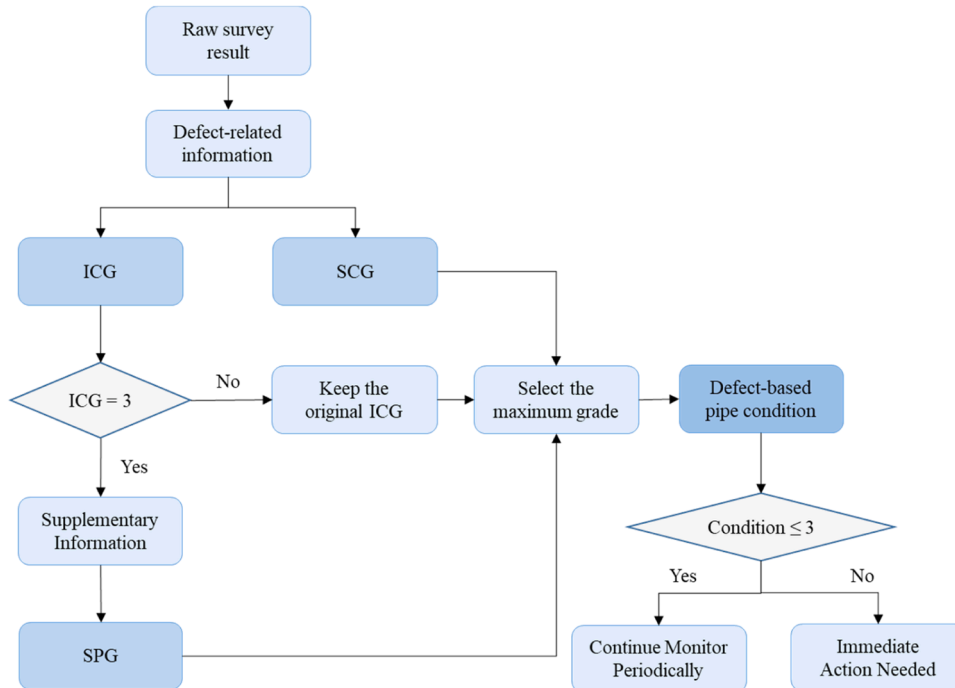


Fig. 2. The process for determining defect-based condition for sewer pipes.

given database. Once structure learning and parameter learning are completed, the predictive inference and data interpretation can be performed given the developed BN model.

2.3. Sensitivity analysis

Sensitivity analysis is used to determine how a target variable is affected by relevant factors, and its essence is to describe the potential probability structure of a model after giving prior probability distributions. Therefore, it can be used to effectively identify IFs in BN models, thereby taking cost-effective measures to reduce risks [27,46]. To obtain a comprehensive assessment, a sensitivity analysis is performed through mutual information (MI), joint probability, and true risk influence (TRI). Besides, two theorems are used to validate the correctness of the constructed BN model. The findings will provide relative authorities with useful suggestions to formulate maintenance strategies.

2.3.1. Mutual information

In this study, a typical sensitive analysis called mutual information (MI) is performed to determine the influence degree of different IFs on the defect-based conditions of sewer pipes. MI is usually used to measure the mutual dependency between two factors, in which the information entropy represents the closeness between two factors [29,42]. The higher the entropy value is, the stronger the correlation represents, and MI can be defined as follows:

$$I(A, B) = H(A) - H(A|B) = \sum_{a \in A} \sum_{b \in B} P(a|b) \log \left(\frac{P(a|b)}{P(a)P(b)} \right) \quad (6)$$

where $I(A, B)$ represents the MI between variables A and B ; $H(A)$ is the entropy of variable A ; $H(A|B)$ presents the entropy of A after giving variable B ; $P(a)$ and $P(b)$ express the marginal probability functions of A and B ; and $P(a|b)$ denotes the joint probability functions of A and B .

2.3.2. Joint probability

Joint probability is another type of sensitivity analysis measuring the joint impacts of different IFs on the target variable. Suppose that A and B represent an IF and target variable, respectively, A_i and B_j represent the i th state of A and j th state of B , respectively, the joint probability of events A_i and B_j can be expressed as:

$$P(A = A_i, B = B_j) = P(A = A_i)P(B = B_j|A = A_i) \quad (7)$$

where $P(A = A_i)$ presents the prior probability of A_i ; $P(B = B_j|A = A_i)$ represents the conditional probability of the B_j occurring when event A_i has already occurred. The higher the joint probability, the greater the influence on the target node.

2.3.3. True risk influence

True risk influence (TRI) is a newly developed approach of sensitivity analysis that can assess the risk impact of IFs under different states [47, 48]. Generally, this metric can be calculated based on the highest risk inference (HRI) and lowest risk inference (LRI) of IFs on the target node in the developed BN models. The first step is to increase each state's probability of the selected IF to 100 %. The second step is to identify two states that produce the highest and lowest probability on the target node, respectively. Then, the difference between the highest and the lowest probability with the original probability should be calculated to obtain HRI and LRI. Finally, the TRI of each IF for each state of target node could be acquired using Eq. (8). The higher the TRI value, the closer the relationship between IF with the target node. Meanwhile, the importance ranking of IF for different statuses of the target node can also be obtained.

$$TRI = \frac{HRI + LRI}{2} \quad (8)$$

2.4. Model validation

To verify the confidence of the developed BN, two axioms should be at least satisfied in the inference process [49,50].

Axiom 1. A slight increase or decrease in the prior probabilities of each test node should contribute to the corresponding increase decrease in the posterior probability of the target node.

Axiom 2. The total influence of the combination of the probability variations of the evidence should be no smaller than the one from a set of risk factors.

In addition, the historical record is also conducted to evaluate if the proposed BN model can reflect the reality by adjusting the parameter setting.

3. Case study

3.1. Data preparation

3.1.1. IF identification from literature

Before collecting necessary data and developing the database, this study firstly identifies a comprehensive list of IFs affecting the defect-based conditions of sewer pipes by reviewing relevant literature, which was obtained on the Web of Science (WoS) using keywords including "sewer pipe condition" and "sewer pipe defect". Then, the content of the retrieved publications was carefully screened, and ultimately 19 articles were selected. By conducting a detailed analysis of adopted IFs in the retrieved publications, 16 frequently used IFs were obtained, as shown in Table 3. To clearly understand the importance of these factors, the frequency of each IF is visualized in Fig. 3. It can be observed that the top six IFs are all related to the physical properties of pipes.

3.1.2. Data collection

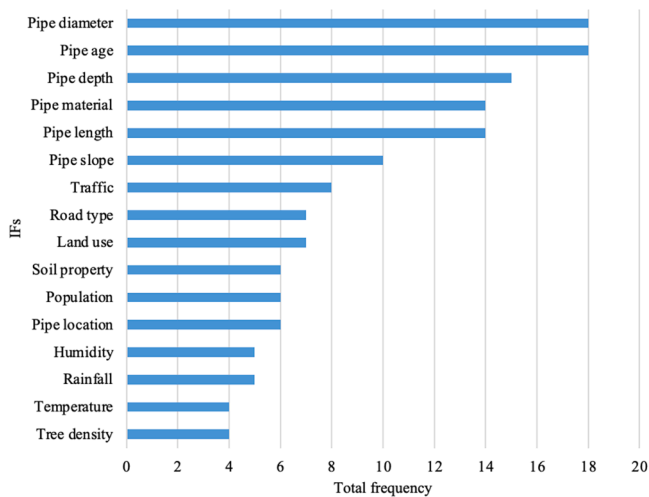
Based on Section 3.1.1, this study collected data from multiple sources to develop an integrated database for sewer pipe condition evaluation in Hong Kong. As an agency dedicated to advanced wastewater and stormwater drainage services, DSD is responsible for operating approximately 4700 km of sewer pipes in Hong Kong, as illustrated in Fig. 4 which provides an overview of Hong Kong's sewer system. Sewer-related data from 2007 to 2021 are available from DSD in two formats. The physical properties of sewer pipes are presented in shapefile (.shp) format in Geographic Information System (GIS), including pipe age, length, diameter, material, shape, slope, location, etc. DSD also provides CCTV reports during the pipe inspection in PDF format, recording the defects detected inside the sewer pipes, including necessary information such as inspection time, defect type, location, etc. Furthermore, the analysis of IFs revealed that the sewer pipe condition is also influenced by the surrounding environment of its laying site, such as population density, land use, traffic, road type and soil properties. Hence, the developed database integrates public census data from the Census and Statistics Department [64]. Meanwhile, to assess the impact of different social activities on the sewer pipe condition, land use (i.e., residential, industrial, and commercial) is taken into account, and collected from the Planning Department [65]. Additionally, road-related data and soil properties (i.e., soil type and soil corrosion) are gathered to evaluate the impacts brought by traffic and soil environment, which are collected from the Transport Department [66] and Civil Engineering and Development Department [67], respectively. Climate variation is also indispensable, such as rainfall, temperature and humidity, which are available in Hong Kong Observatory [68] according to the statistics of each weather station.

Table 3

Sources of IFs based on the retrieved literature.

Refs.	IFs from the retrieved literature															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Tran et al. [51]	✓	✓			✓	✓	✓					✓	✓			✓
Tran et al. [52]	✓	✓			✓	✓	✓					✓	✓			✓
Salman [53]	✓	✓	✓	✓	✓	✓					✓					
Salman and Salem [54]	✓	✓	✓	✓	✓	✓					✓					
Ennaouri and Fuamba [55]	✓	✓		✓	✓			✓					✓			
Harvey and McBean [56]	✓	✓	✓	✓	✓	✓			✓	✓						
Kimutai et al. [57]	✓	✓	✓	✓	✓									✓	✓	
Bakry et al. [58]	✓	✓	✓	✓	✓			✓		✓	✓		✓			
Bakry et al. [59]	✓	✓	✓	✓	✓			✓		✓	✓		✓			
Anbari et al. [60]	✓	✓		✓	✓		✓					✓				
Alsaqqar et al. [61]	✓	✓	✓	✓	✓	✓		✓								
Elmasry et al. [5]	✓	✓	✓	✓	✓		✓									
Kaddoura et al. [20]	✓	✓	✓	✓	✓					✓	✓		✓			
Khudair et al. [62]	✓	✓	✓	✓	✓	✓		✓								
Yin et al. [16]	✓	✓			✓			✓	✓	✓		✓				
Lee et al. [63]	✓	✓	✓		✓	✓	✓		✓	✓	✓			✓		
Fontecha et al. [17]	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓			✓	✓	✓
Abdelkhalek and Zayed [1]	✓	✓	✓	✓	✓	✓		✓	✓	✓				✓	✓	✓
Ma et al. [6]	✓	✓	✓	✓	✓			✓	✓	✓				✓	✓	✓
Total	18	18	14	14	15	10	6	8	6	7	7	4	6	5	4	5

Note: 1. Pipe age; 2. Pipe diameter; 3. Pipe length; 4. Pipe material; 5. Pipe depth; 6. Pipe slope; 7. Pipe location; 8. Traffic; 9. Population; 10. Land use; 11. Road type; 12. Tree density; 13. Soil property; 14. Rainfall; 15. Temperature; 16. Humidity.

**Fig. 3.** Frequency of summarized IFs in the retrieved literature.

3.1.3. Data processing

After collecting the original data, data processing is further conducted, with the detailed procedures shown below:

Step 1. Data integration

Integration of data from multiple sources forms the basis of this research. At this stage, data extracted from CCTV reports is linked into the GIS platform by conducting the connection operation based on the unique pipe ID. Afterwards, environment-related factors and climate-related factors are matched to pipes by spatial join techniques in ArcGIS software. The integration process of climate-related factors is cumbersome, in which each pipe is firstly linked to the corresponding weather station based on its geographical location and then matched with publicly available data collected by the HKO. This step considers the inspection date of defects and match data on this basis.

Step 2. Data cleansing

Generally, inaccurate data may negatively impact the model

development process. In this study, data cleansing is carried out in two sub-steps. Firstly, duplicated pipe records are consolidated since each defect observed during the inspection is considered as a separate record, which could successfully compress the database and preserve the integrity of the original data. Secondly, records with too much missing information related to the IFs identified in Section 3.1.1 are removed. Finally, 23,000 sewer pipe records are reserved after the data cleansing.

Step 3. Determination of IFs and target variable

The cleansed database in the previous step needs to be processed, where IFs with continuous features need to be discretized, such as pipe age, length, diameter, population, traffic, etc.; Meanwhile, states of some IFs need to be simplified to reduce data complexity when quantifying interdependencies. To comprehend the characteristics of the processed database, a statistical analysis and categorization of all IFs have been presented in Table 4. The database reveals substantial variabilities in pipe characteristics, exemplified by pipe age, which spans from 1 to 116 years with an average of 37.55 years; meanwhile, the average values of pipe length and diameter are 18.25 m and 420.23 mm, illustrating considerable variabilities in service longevity and the diverse specifications of the pipes under study. Furthermore, traffic volume exhibits a substantial range from 302.78 to 158,465, while population density similarly varies between 6014 and 26,271, with an average of 16,804.62, thereby highlighting significant regional disparities. Additionally, factors such as average temperature (23.48 °C) and humidity (77.68 %) further reflect the climate characteristics of Hong Kong.

Afterwards, the condition of each pipe needs to be inferred based on their defect records according to the steps mentioned in Section 2.1.3., in which the proportion of each defect type is illustrated in Fig. 5. It can be observed that deposits make up the largest proportion (20.19 %). The second most common defect is fracture, accounting for 13.84 %, which is followed by joint displaced and crack, with percentages of 13.39 % and 12.09 %, respectively. In contrast, repair, junction and connection defective constitute the smallest part, occupying only 0.51 % in total. Based on the defect records, the ICG, SCG and condition of each pipe can be determined. Fig. 6 illustrates the distribution of sewer pipes across various categories.

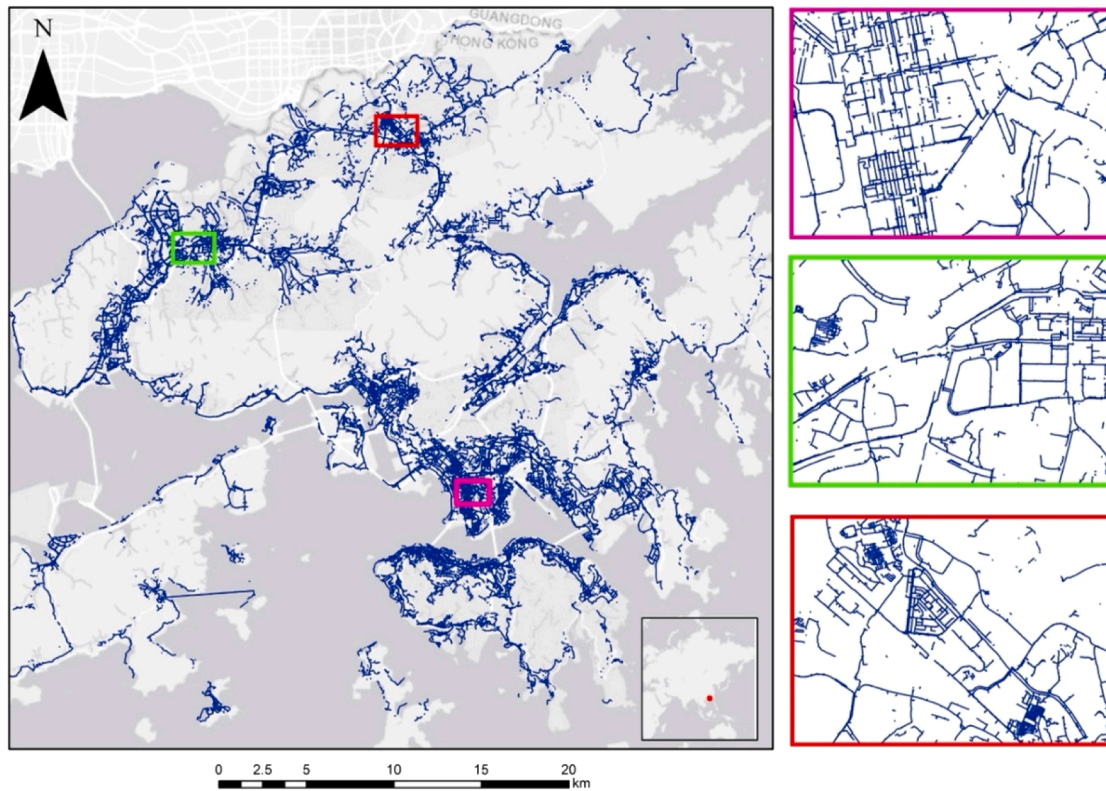


Fig. 4. Overview of the sewer system network in Hong Kong.

3.2. Model development

The structure of the BN model is learned through Python and visualized by *GeNIe* software. As shown in Fig. 7, the trained BN structure includes 15 nodes and 31 links. Pipe physical-related factors, environment-related factors, and climate-related factors are represented by blue, green, and yellow nodes, respectively. The links explore the relationship among pairs of IFs and between each IF and pipe condition, which could express useful information on the causation analysis of defect-based pipe conditions. For instance, the node of the pipe condition is defined as the child node of three parent nodes (pipe age, pipe diameter and population), which means that these parent nodes potentially affect the pipe condition.

The probability distributions for all nodes are also calculated during the structure learning process, and the results are presented in Fig. 8. It can be found that among the active sewer pipes in Hong Kong, most of them are in perfect condition, accounting for about 47 % of the total. From the perspective of physical properties, the number of pipes older than 60 years is minimal, accounting for only 7 %. Meanwhile, about 41 % of the pipes are 10 meters or less in length, and nearly half have a diameter between 200 mm and 400 mm. As for the pipe material, most pipes are made of concrete and vitrified clay, accounting for 54 % and 39 %, respectively. Besides, about 94 % of pipes are located in urban areas. Regarding environment-related factors, about 42 % of pipes are in residential areas, and about 65 % are located in areas where the annual average daily traffic volume is less than or equal to 20,000. The population distribution is relatively even, among which the majority of pipes are situated in areas with a population density of 15,000 to 17,000, taking up about 23 %. The proportions of roadways and sidewalks in road type are similar at 48 % and 44 %, respectively. As to soil properties, granitic rock is the most common soil type taking up 42 % of the total, and about 65 % of pipes are in moderately corrosive soils. In terms of climate factors, the climatic conditions of most pipes involved in the database are higher than 25 degrees Celsius in temperature (48 %), less

than 100 mm in rainfall (67 %), and between 75 % and 85 % in humidity (59 %).

4. Results and discussion

4.1. Sensitivity analysis

4.1.1. Mutual information

The MI value of IFs is calculated to prioritize the influence magnitude to the target node, where the higher value corresponds to the greater impact. The MI value and corresponding percentage for all IFs in the BN are listed in Table 5. It can be observed that “pipe age” has the greatest impact on the “pipe condition” with a MI value of 0.124, which is followed by “pipe diameter” with a MI value of 0.032. Next, “population” and “soil type” rank the third and fourth, with MI values of 0.027 and 0.024, respectively. In addition, the MI values of the remaining IFs are lower than 0.02, and their corresponding perceptions are also less than 1 %. Therefore, the top four influential IFs are selected as judgment criteria to analyse and validate the BN model in further steps.

4.1.2. Joint probability

After ranking the importance of all IFs based on MI values, an additional sensitivity method is applied to investigate the detailed impact of the most influential IFs on “pipe condition”. The joint probability between IFs and the target node can be obtained by sequentially adjusting each state of each factor to 100 %. The probability results of each state about different factors are shown in Table 6, where the first row presents the initial probabilities distribution of “pipe condition”, which allows a comparative display of how the probability of each condition changes when the selected influential factor is set in a specific state. To illustrate the different influence of factors on each pipe condition, the maximum and minimum impacts are marked in bold. Thus, some valuable insights can be obtained from the new findings. For instance, pipes with an age less than or equal to 20 years, a diameter

Table 4
Categorization and statistical description of the IFs used in this study.

Number	IFs	Type	Unit	Min.	Max.	Mean	Std.	States
<i>Pipe physical-related factors</i>								
1	Pipe age	Continuous	year	1	116	37.55	16.21	Less than or equal to 20, Between 20 and 30, Between 30 and 40, Between 40 and 50, Between 50 and 60, Large than 60
2	Pipe length	Continuous	m	0.10	245.34	18.35	16.71	Less than or equal to 10, Between 10 and 20, Between 20 and 30, Between 30 and 40, Between 40 and 60, Large than 60
3	Pipe diameter	Continuous	mm	75	3000	420.23	349.01	Less than or equal to 200, between 200 and 400, between 400 and 600, between 600 and 800, between 800 and 1000, Large than 1000
4	Pipe material	Categorical	–					Concrete, Vitrified Clay, Others
5	Pipe location	Categorical	–					Rural, Urban, Waterbody
<i>Environment-related factors</i>								
6	Population	Categorical	–	6014	26,271	16,804.62	3192.82	Less than or equal to 15,000, Between 13,000 and 15,000, Between 15,000 and 17,000, Between 17,000 and 19,000, Between 19,000 and 21,000, Large than 21,000
7	Traffic	Categorical	AADT	302.78	158,465	21,444.52	23,068.14	Less than or equal to 10,000, Between 10,000 and 20,000, Between 20,000 and 30,000, Between 30,000 and 40,000, Large than 40,000
8	Land use	Continuous	–					Other specified uses, Commercial, Residential, Comprehensive development area, Government, Institution or community, Open space
9	Road type	Categorical	–					Carriageway, Footway, Others
10	Soil corrosion	Categorical	–					Noncorrosive, Mildly corrosive, Highly corrosive
11	Soil type	Categorical	–					Fill, Granitic rocks, Superficial deposits, Tuff and lava
<i>Climate-related factors</i>								
12	Rainfall	Continuous	mm	0.22	1332.00	124.15	158.52	Less than or equal to 100, Between 100 and 200, Between 200 and 300, Between 300 and 400, Large than 400
13	Humidity	Continuous	%	53.00	99.00	77.68	7.85	Less than or equal to 70, Between 70 and 75, Between 75 and 80, Between 80 and 85, Large than 85
14	Temperature	Continuous	°C	10.70	30.8	23.48	4.85	Less than or equal to 15, Between 15 and 20, Between 20 and 25, Large than 25

greater than 1000 mm, laid in areas with a population density between 19,000 and 21,000, and surrounding soil types of superficial deposits, are least likely to deteriorate to “critical” condition. In contrast, pipes are most likely to be in “critical” condition when their age is between 50 years and 60 years, the diameter is less than or equal to 200 mm, the laying site has a population density of less than or equal to 15,000, and the surrounding soil type is granitic rocks.

4.1.3. True risk influence

Based on the initial probability and result of joint probability in

Table 6, the TRI values of the top four IFs can be obtained. As shown in Table 7, an example is given to display the calculation process of the TRI value of “pipe age” against “perfect” condition. It can be observed that when each state of “pipe age” reaches 100 % in turn, “pipe age” between 20 and 30 years has the greatest impact on “perfect” condition, while “pipe age” greater than 60 years has the least impact. Under these two scenarios, the probability of the “perfect” condition is 52.77(%) and 38.45(%), respectively. Therefore, the value of HRI and LRI is the difference between these two values and initial probability (46.54), which are 6.23(%) and 8.09(%), respectively. Then, the TRI value of “pipe

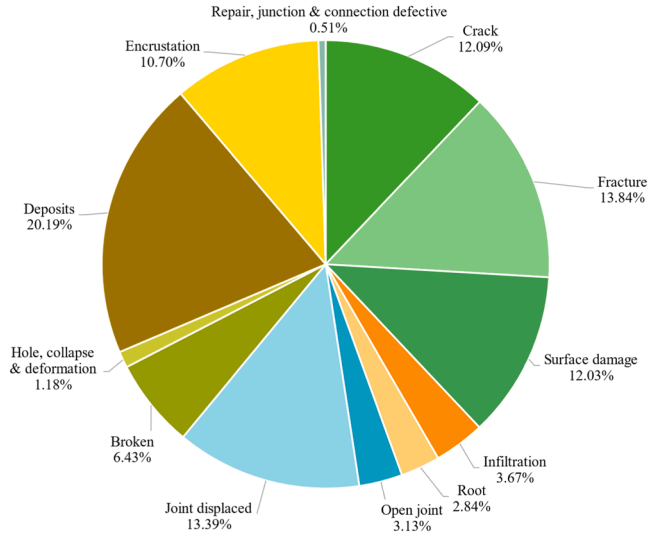


Fig. 5. Proportion of various types of defects in Hong Kong's sewer network.

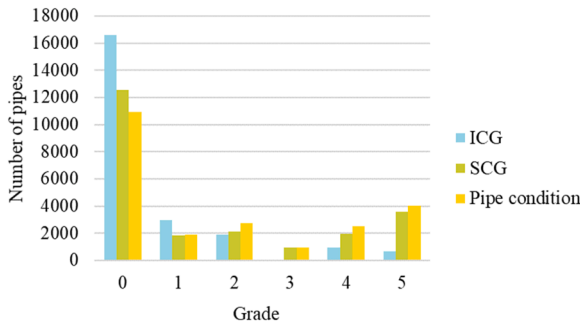


Fig. 6. The number of sewer pipes with different grades.

age" for "perfect" condition can be calculated according to Eq. (8), which is 7.16. The same calculation process can be applied to get TRI between other factors and each pipe condition, the final results are

presented in Table 8.

It is obvious that the impact of IFs on pipe condition varies with the different states. According to the TRI values, the importance of IFs for six pipe conditions is ranked from 1 (the most influential) to 4 (the least influential), as illustrated in Fig. 9. It is apparent that "pipe age" has greater impacts on "fair", "poor" and "critical" conditions than others. The "pipe diameter" presents a stronger relationship with "perfect" condition, while "population" exerts a greater influence on "excellent" and "good" condition. Besides, the impact of "soil type" on "pipe condition" is relatively lower than other IFs.

4.2. Model validation

In addition to sensitivity analysis of individual IFs, the comprehensive impact of multiple IFs is also used to validate the model. When taking the top four IFs as a set of variables, the probability change of the target node can be observed by slightly adjusting their prior probabilities in sequence, where the change rate is set to be 5 %. The "pipe age" is selected as the first node, in which the state that has the highest impact on the "perfect" state in "pipe condition" is increased by 5 %, meanwhile, the state that has the lowest impact on "perfect" state is decreased by 5 %. This process is presented as "~5 %" in Table 9. Then, the same procedure is applied to the remaining IFs in sequence, and the corresponding values of comprehensive change are recorded in Table 9. Similarly, the same parameter adjustment process is used for the other states of the target node until all of them are traversed.

According to Tables 6 and 9, changes in the prior probability of IFs will cause corresponding changes in the posterior probability of the target node, which shows that the proposed BN model is consistent with Axiom 1. In addition, it can be observed from Table 9 that the probability of the target node increases with the continuous update of the probability distribution of IFs, which proves that the model satisfies Axiom 2. Therefore, when both Axiom 1 and Axiom 2 are satisfied, the correctness of the proposed model is verified.

Furthermore, a historical pipe record that isn't contained in the database during BN model training is selected to further evaluate the effectiveness of the model. The pipe "SWD703XXX2" was inspected in April 2019 and is located at Murray Road. During the inspection, it was discovered that there were two medium encrustations at the pipe. Therefore, the pipe condition is determined as poor. According to the

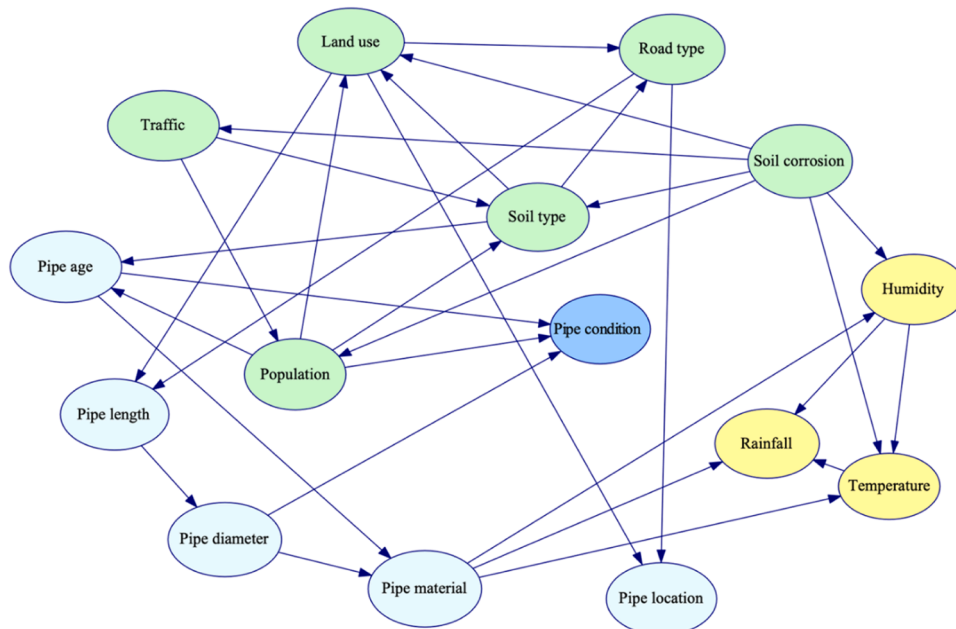


Fig. 7. The structure of the BN model.

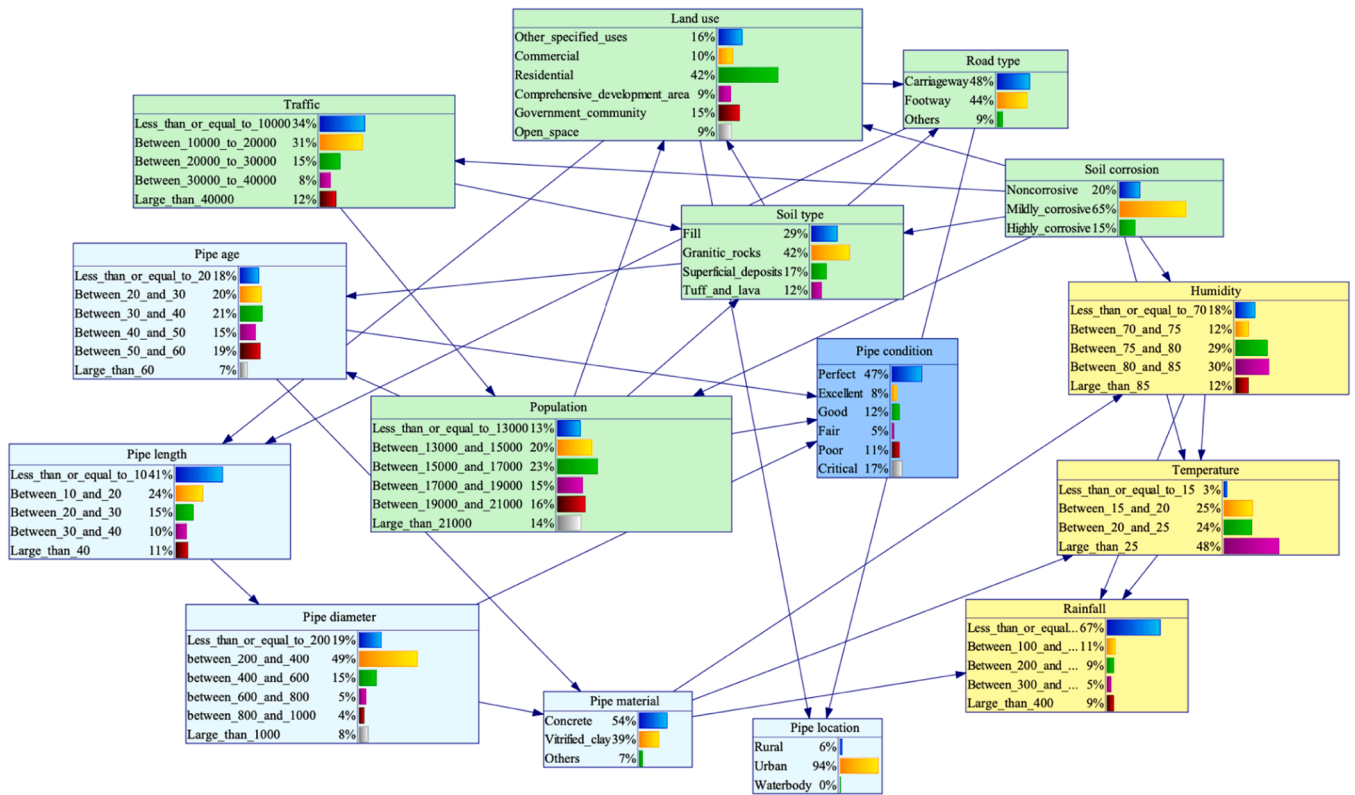


Fig. 8. The probability distribution of the BN model.

Table 5
Mutual information between pipe condition and IFs.

Node	Mutual information	Percentage (%)
Pipe condition	1.495	100.00
Pipe age	0.124	8.305
Pipe diameter	0.032	2.173
Population	0.027	1.797
Soil type	0.024	1.613
Pipe material	0.014	0.926
Land use	0.013	0.863
Pipe location	0.008	0.561
Pipe length	0.007	0.493
Road type	0.006	0.429
Traffic	0.006	0.373
Humidity	0.005	0.332
Rainfall	0.040	0.289
Temperature	0.003	0.208
Soil corrosion	0.002	0.147

pipe record of CCTV reports, parameter setting of the IFs are presented in Table 10, in which climate-related factors are unknown. The real case is simulated with the proposed BN model, as illustrated in Fig. 10. It can be seen that the probability of “poor” state is the highest at 85 %, which proves the validity of the developed BN model. Thus, the model can be used to justify the pipe condition when CCTV or other technology is limited.

4.3. Scenario analysis

Once the BN model is trained and validated, it can be used to explore the influence between related variables in potential scenarios. Technically, the scenario analysis of Bayesian networks is performed by adjusting the state of one or more nodes, during which new evidence is propagated through the developed system to infer changes in the target variable [69]. Therefore, the analysis of relevant scenarios can reveal

the possible pipe conditions, thereby effectively helping authorities formulate targeted maintenance measures.

4.3.1. Scenario one: the most likely scenario for “fair” condition

Since “fair” is a special state before pipe failure, the first scenario is designed to investigate the most likely scenario for “fair” condition, which needs to adjust the state of “fair” to 100 %. As shown in Fig. 11, the probability distribution of some nodes has changed significantly, such as the less than or equal to 20 and between 20 and 30 in “pipe age”, less than or equal to 200 and between 200 and 400 in “pipe diameter”, vitrified clay in “pipe material”, and superficial deposits in “soil type”. This suggests that pipes in these scenarios need more attention, since “fair” condition acts as a tipping point, leading to an increased likelihood of pipe deterioration, especially as the pipe ages. Therefore, authorities should increase the frequency of inspections and enhance routine maintenance of such pipes.

4.3.2. Scenario two: the combined impact of important IFs

The combined influence of the top four important IFs (pipe age, pipe diameter, population and soil type) on the target state “critical” can be assumed as another scenario, since “critical” state implies that the pipe has failed. When the probability of the state that generates the highest joint probability with “critical” in each IF is set to 100 %, the most extreme scenario when the pipe condition is “critical” can be presented. As illustrated in Fig. 12, the probability of state “critical” increased from 17 % to 72 %. Thus, relevant departments should take preventive measures against these key factors, such as enhanced monitoring and assessment of pipes that have been in service for more than 50 years and whose diameter is less than 200 mm. Meanwhile, the health of pipes in less densely populated areas shouldn’t be ignored.

4.3.3. Scenario three: the possible scenario in extreme weather

Hong Kong is situated in a subtropical climate zone and is frequently affected by tropical cyclones from the Pacific Ocean and the South China

Table 6
Joint probability of the BN model.

Pipe condition	Perfect	Excellent	Good	Fair	Poor	Critical
Initial value	46.54	8.34	12.10	4.55	11.42	17.05
Pipe age						
Less than or equal to 20	52.47	12.72	13.46	10.21	8.38	2.76
Between 20 and 30	52.77	8.02	11.13	12.29	12.43	3.37
Between 30 and 40	45.98	7.24	12.36	0.30	18.15	15.97
Between 40 and 50	44.78	6.42	10.75	0.26	18.74	19.05
Between 50 and 60	39.20	6.96	12.55	0.34	3.37	37.58
Large than 60	38.45	9.16	12.31	0.75	2.35	36.97
Pipe diameter						
Less than or equal to 200	39.51	5.77	8.44	5.73	15.95	24.61
between 200 and 400	43.67	7.52	12.43	5.15	11.76	19.46
between 400 and 600	51.84	10.77	13.38	3.49	8.62	11.89
between 600 and 800	56.49	11.96	13.10	2.06	8.32	8.06
between 800 and 1000	53.66	11.44	15.40	3.28	8.80	7.43
Large than 1000	61.35	11.07	14.04	2.23	6.98	4.32
Population						
Less than or equal to 15,000	48.63	2.08	4.99	5.26	14.95	24.10
Between 13,000 and 15,000	50.50	5.41	9.05	4.04	12.37	18.63
Between 15,000 and 17,000	44.34	10.14	13.61	4.29	11.05	16.58
Between 17,000 and 19,000	44.02	9.20	12.97	4.94	11.05	17.82
Between 19,000 and 21,000	50.75	10.36	11.41	5.53	9.84	12.12
Large than 21,000	40.38	12.32	20.71	3.52	9.45	13.61
Soil type						
Fill	46.67	7.38	11.46	4.54	12.09	17.86
Granitic rocks	45.16	7.94	12.04	3.46	11.28	20.12
Superficial deposits	47.99	10.18	13.50	6.21	10.77	11.34
Tuff and lava	49.14	9.46	11.84	6.09	11.21	12.26

Table 7
An example of TRI about “pipe age” for “perfect” condition.

Pipe age							
Less than or equal to 20	/	100 %	0	0	0	0	0
Between 20 and 30	/	0	100 %	0	0	0	0
Between 30 and 40	/	0	0	100 %	0	0	0
Between 40 and 50	/	0	0	0	100 %	0	0
Between 50 and 60	/	0	0	0	0	100 %	0
Large than 60	/	0	0	0	0	0	100 %
Perfect	46.54	52.47	52.77	45.98	44.78	39.20	38.45
HRI	6.23						
LRI	8.09						
TRI	7.16						

Table 8
TRI of IFs for all pipe conditions.

	Perfect	Excellent	Good	Fair	Poor	Critical	Average
Pipe age	7.16	3.15	1.35	6.02	8.20	17.41	7.21
Pipe diameter	10.92	3.10	3.48	1.83	4.48	10.14	5.66
Population	5.18	5.12	7.86	1.00	2.75	5.99	4.65
Soil type	1.99	1.40	1.02	1.37	0.66	4.39	1.81

Sea. As an extreme weather phenomenon, cyclones bring about impacts such as strong winds, heavy rainfall, storm surges, and sea waves. Among these, heavy rainfall can lead to substantial inflows of sewage carrying debris, silt, and other materials into the sewer system, potentially impairing its functionality. On June 6, 2020, influenced by a low-pressure trough, Hong Kong experienced continuous rainfall for seven hours, with precipitation exceeding 200 mm in multiple areas (see Fig. 13). Concurrently, the HKO issued a Black Rainstorm Warning Signal. This study selected “Ting Kok Road”, one of the flooding black spots identified by the Hong Kong government for scenario simulation, to observe the distribution of pipe conditions in this area under extreme rainfall. According to meteorological data, the rainfall in this area reached 445 mm during the Black Rainstorm Warning Signal. Additionally, known variables for this area include a population density of 19,459, a traffic volume of approximately 12,700, soil type classified as tuff and lava, and moderate soil corrosion. By inputting the known data into the constructed Bayesian network, the simulation results are presented in Fig. 14. The results indicate that approximately 72 % of the pipes only require regular monitoring. Among the remaining pipes, 6 %

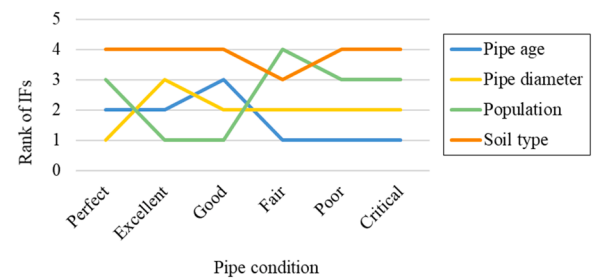


Fig. 9. The top four important IFs for all pipe conditions.

Table 9
The result of minor changes in IFs.

	/	~5 %	~5 %	~5 %	~5 %
Pipe age	/	/	~5 %	~5 %	~5 %
Pipe diameter	/	/	~5 %	~5 %	~5 %
Population	/	/	/	~5 %	~5 %
Soil type	/	/	/	/	~5 %
Perfect	46.54	47.21	48.42	48.95	49.14
Excellent	8.34	8.69	8.98	9.53	9.66
Good	12.10	12.3	12.5	13.33	13.4
Fair	4.55	5.15	5.43	5.54	5.68
Poor	11.42	12.28	12.72	13.08	13.16
Critical	17.05	18.93	20.07	20.75	21.24

Table 10
The parameter setting of the real case verification.

IFs	State	IFs	State
Pipe age	46 years	Traffic	57,657.78
Pipe length	37.98 m	Road type	Carriageway
Pipe diameter	760 mm	Soil corrosion	Mildly corrosive
Pipe material	Others	Soil type	Fill
Pipe location	Urban	Rainfall	N/A
Population	12,501	Humidity	N/A
Land use	Open space	Temperature	N/A

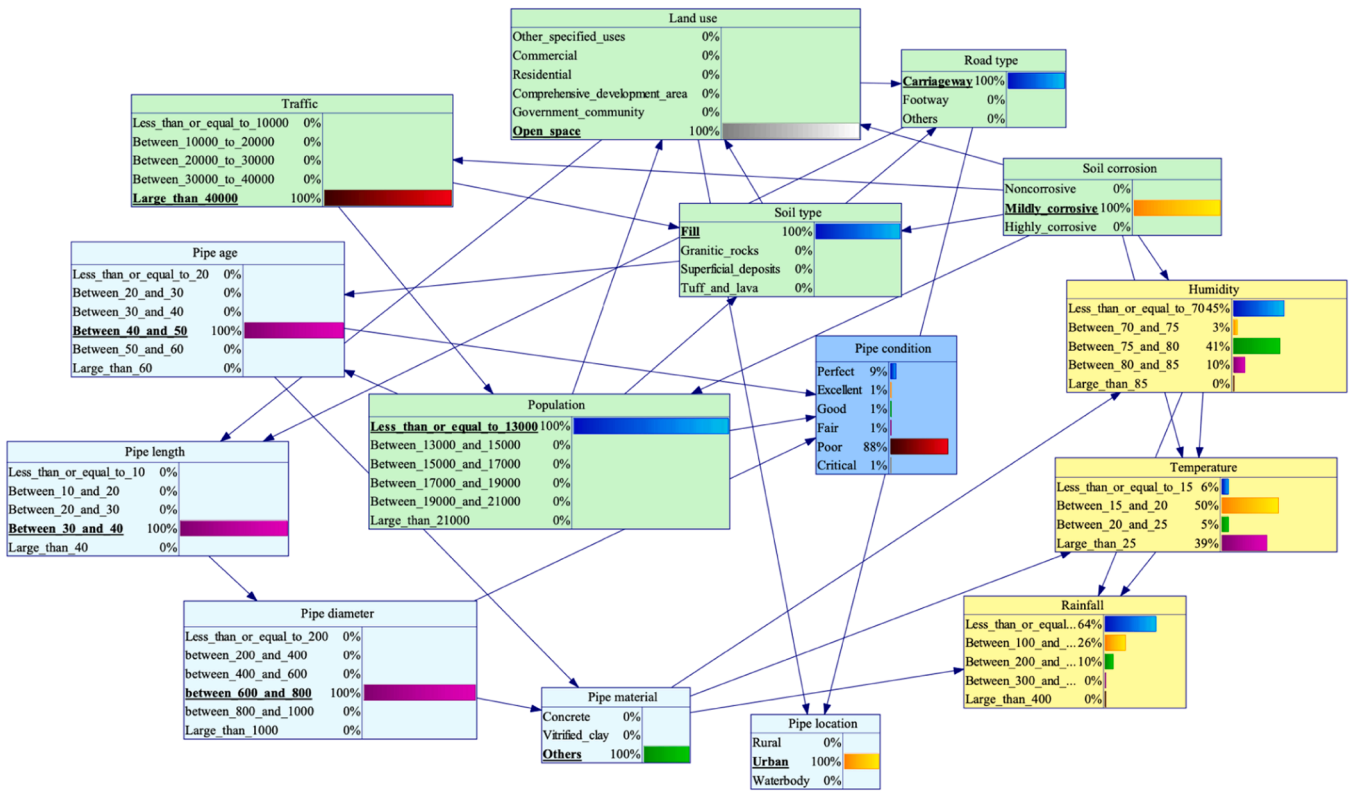


Fig. 10. Real case verification of the BN model.

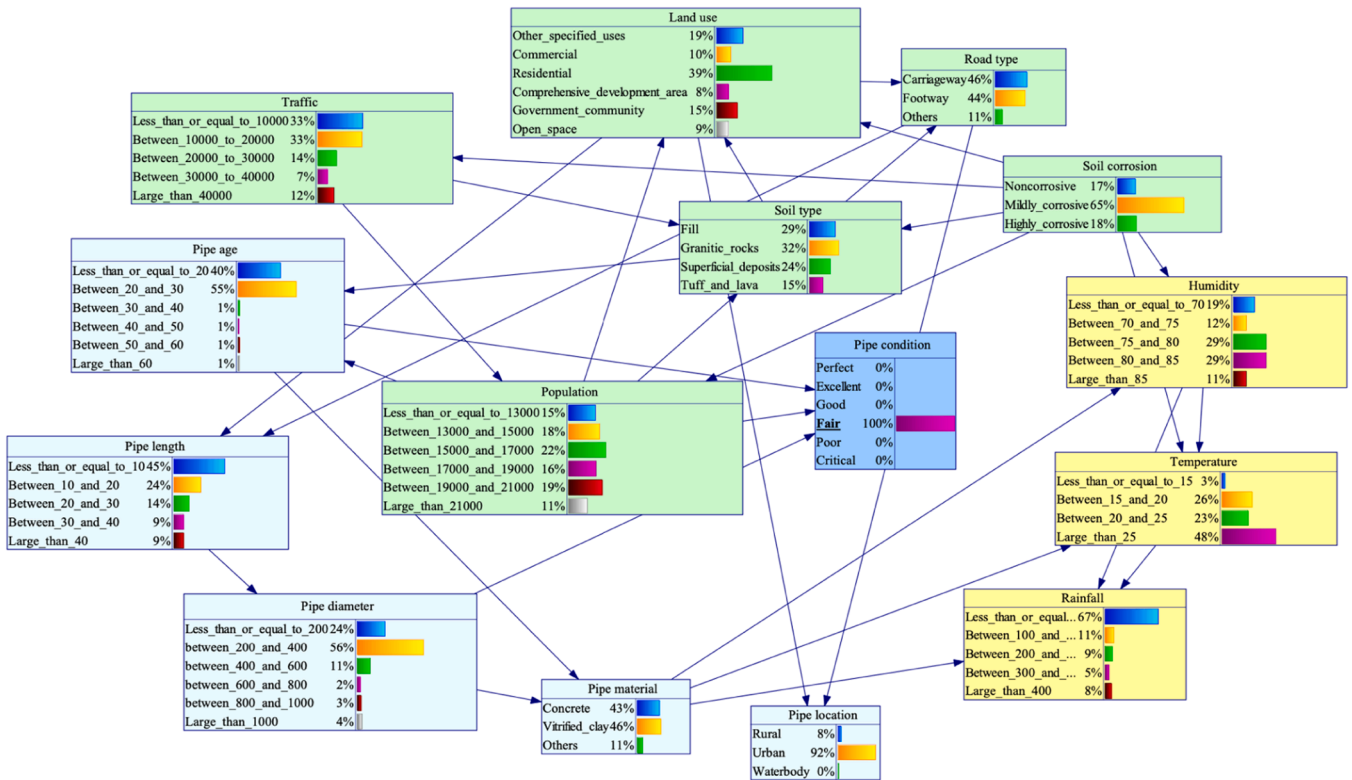


Fig. 11. The most likely scenario for "fair" condition.

are in fair condition, necessitating more frequent monitoring during routine maintenance. The other 21 % are potentially in poor or critical state, requiring immediate repair measures.

5. Conclusions

This paper develops a data-driven Bayesian network (BN) model to

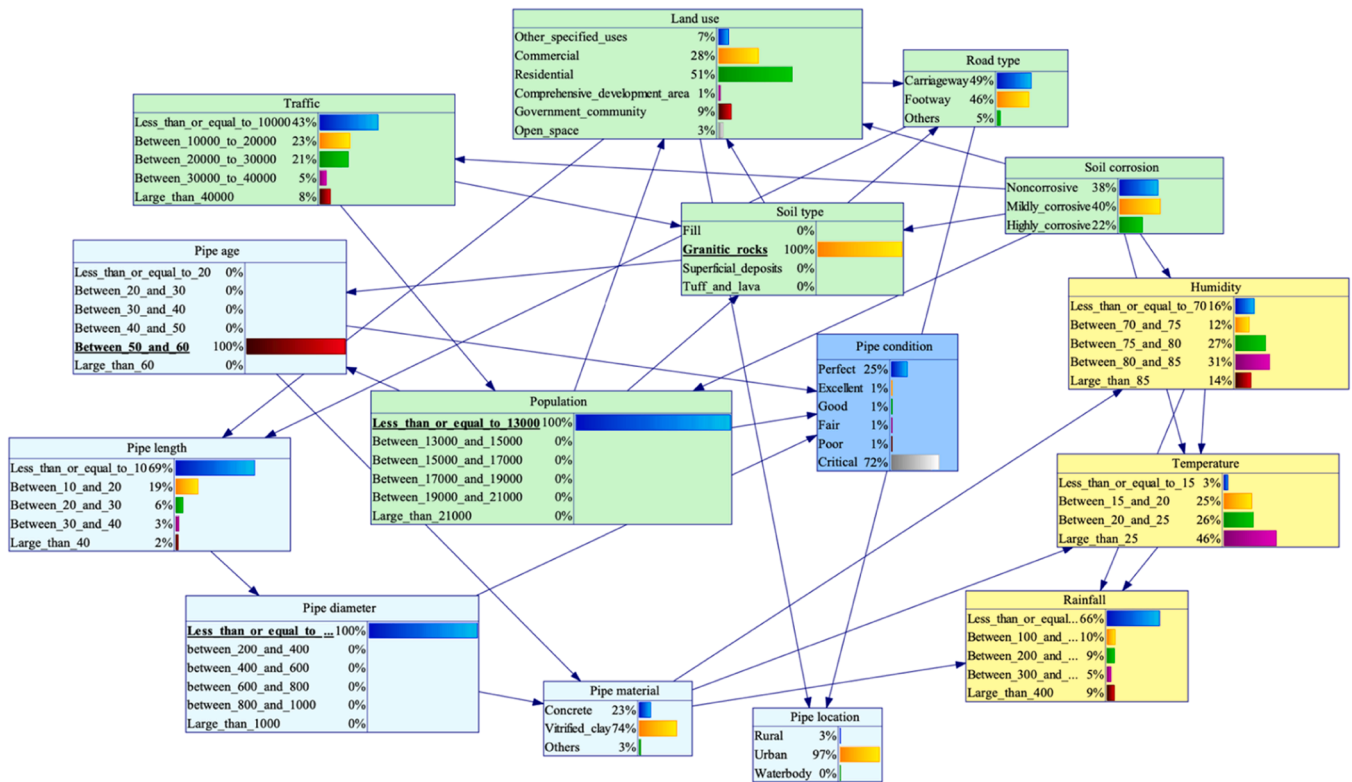


Fig. 12. The combined impact of important IFs for “critical” condition.

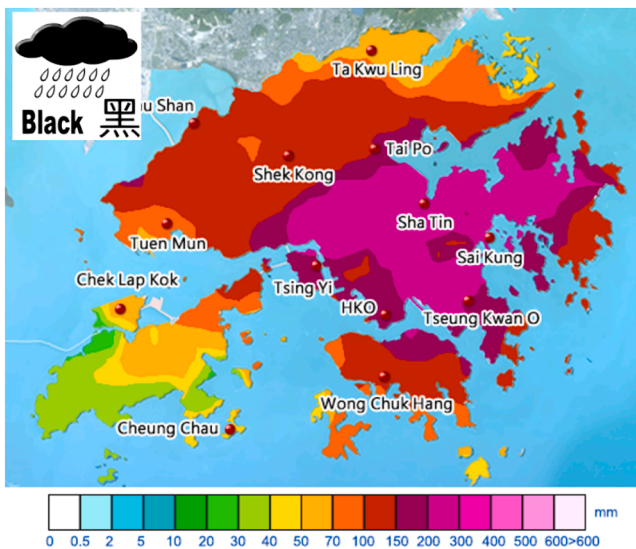


Fig. 13. Rainfall distribution map on June 6, 2020, collected from HKO.

analyze the impacts of different IFs on sewer pipes' defect-based conditions and their intrinsic relationships. Firstly, reviewing existing literature and evaluation protocols related to pipe defects and conditions helps identify a series of factors that may affect sewer pipe's defect-based conditions. Secondly, taking Hong Kong as a research case, this study establishes a comprehensive database containing fourteen IFs, which integrate multiple data sources related to sewer pipes, covering multiple types of factors such as physics, environment and climate. Then, a data-driven Bayesian network model is created to evaluate the probability distribution of sewer pipe conditions, which could clarify the interrelationship between each IF and the defect-based condition of sewer pipes. Finally, sensitivity analysis, model validation and scenario

analysis are used to identify critical IFs and evaluate the model's reliability under different scenarios.

This study enhances the sewer management system by proposing an aggregated index of pipe condition that integrates CCTV-reported defect type and frequency with different grading indicators. After determining a comprehensive list of IFs that affect the condition of sewer pipes based on past literature and completing the multi-source database development, this study constructed the most reasonable model structure through the score-based Bayesian searching algorithm, which effectively avoided the subjectivity that may be caused by expert evaluation. Furthermore, after exploring the impact of different IFs on the condition of sewer pipes, as well as their complex interrelationship, this study classified the sewer pipe conditions into six levels. This refined status classification enables us to understand the defects in sewer pipes better, which facilitates the proposal of targeted interventions and cost-effective maintenance strategies more precisely. Therefore, this model can not only be employed to study the potential condition of a single sewer pipe but also to evaluate the probability distribution of pipe conditions at the regional scale.

The results provide valuable implications for the defect-based condition assessment of sewer pipes. In particular, pipe age, pipe diameter, population and soil type are the top four IFs, among which pipes with age greater than 50 years, diameter less than 200 mm, population less than 15,000, and located in fill and granitic rocks are more likely to be in dangerous conditions (“poor” and “critical”) with more serious defects (e.g., multiple fracture, broken, collapse). Meanwhile, the precision of the proposed model can achieve 85 % in real case study. It is recommended that relevant departments shall strengthen monitoring or appropriately adjust pipe types under such high-risk scenarios. In addition, pipes in the “fair” state also deserve more attention. As an important node for a pipe switching from non-failure to failure, many situations may aggravate pipe deterioration, such as changes in the soil environment, special geographical features, previous failures, heavy traffic load, etc. Thus, adjusting the frequency of regular inspections based on pipe conditions and formulating proactive maintenance

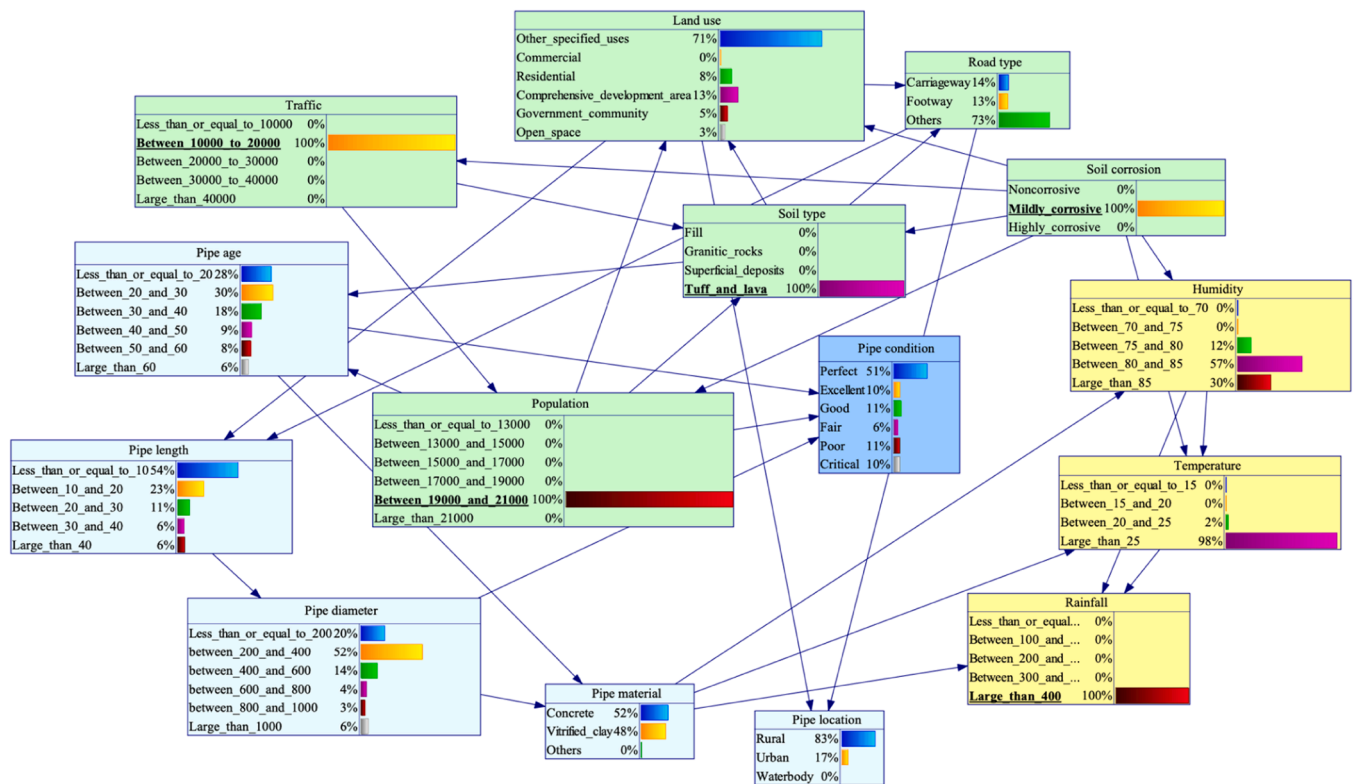


Fig. 14. The possible scenario in extreme weather.

measures might effectively avoid the occurrence of pipe failure.

The comprehensive database and the well-developed BN model in this paper are of great significance in the practice of sewer pipe assessment and can be used as effective tools for stakeholders in the field. Despite limitations, such as limited data volume and model accuracy, this study also lays the foundation for future research. In future studies, model development algorithms can be continuously updated and compared to improve model accuracy. At the same time, database expansion is also a part worth focusing on, including increasing the amount and type of data, e.g., collecting sewer pipe data from different regions and incorporating socio-economic factors, which can enhance the robustness of the model and make it more accurately adaptable to condition analysis of sewer pipes under different scenarios.

CRedit authorship contribution statement

Shihui Ma: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Tarek Zayed:** Writing – review & editing, Supervision, Funding acquisition. **Jiduo Xing:** Writing – original draft, Methodology, Conceptualization. **Zhihao Ren:** Writing – original draft, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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